Linear data Structures : →

① Stacks, Queues, Arrays
② Linked Lists
↳ Singly → Forward Traversal (H → T)
↳ Doubly → Both Side "
↳ Circular

T⟷H
DEQ

Array → index | 2 | 9 | 1 | 3 |

4th arr[3] → TC → O(1)
(n − 1) → index

Array
| 2 | 9 | 6 | 1 | 4 |

3rd = n − 1
arr[2] index
Search complexity
TC = O(1)

Linked List
nodes
1 → 2 → 3 → null
address head tail address

nullptr

3rd node TC(n)
Search Complexity

Insertion Time Complexity :

arr | 1 | 2 | 4 | 5 | 6 | 7 |
③ O(n)
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | |
n steps

O(1) TC(O(n))
Access data ⇒ Array

Linked List
1 → 3 → 4 → null
2
TC O(1)
Store ⇒ LL

Node * node
Forward Traversal

| 20 | 1000 | → | 30 | 2000 | → | 40 | | → null
data next (Node) data next (Node) data next

head → 1000 1000 tail 2000

```
Struct Node {
public int data;
       Node* next;
}
```

```
class Node {
private int data
        Node * next;
}
```

Operations in a linked list :

① Insert → ⓘ Insert at Head
ⓘⓘ Insert at Tail
ⓘⓘⓘ Insert after Specific

② Deletion → ⓘ Delete Head
ⓘⓘ Delete Tail
ⓘⓘⓘ Delete Specific

③ Display ① → ② → ③ → null
node → node → node node
1 → 2 → 3

1 head 2 second tail 3 third → null

Insert At Head :

iah (head, nd) {
}

(n) 1 → 2 → 3 → null
(h) → pointer **

newdata ⓪

```
Node* node = new Node ( );
    node ->data = newdata;
    node.next = h
    h = node
```

**head = 1

Insert At Tail :

Create a new Node
new Node.next = null

① Empty : [head = = nullptr]
make the new node as head

② Only one node :
h
① → null
t

```
if (head.next == null)
    h.next = nNode;
```

③ Multiple nodes:

l = l.next
while (l != null)

l.next = nN
1 → 2 → 3 → null
last last last last
l = h ④ nN

while (last.next != null)

Steps: Head :

① Create Node → new data
② Add data to the node
③ Point next of new node → head
④ Point head to new node

nN ⓪ ② A → C → X
① ② ③
h ① h

① ② ③
⓪

P.n
h 1 → 2 → 3 → null
t
nN.n = p.n
p.n = nn

p → next = ②
nN → next = p.next

New Node — ①

| A | C | M | A | → n

```
if (prev == null)
  cout << " cannot ";
  return ;
```

nn.n = p.n
p.n =

S → R
null

1 → 2 → ③ → 4
insert after target
pointer (head, 1)